MICROSOFT 365 STORAGE

Tenant Storage Assessment Script

Version 2.4 Usage and Sharing Guide

> **The headline feature:** The script creates a polished, standalone HTML assessment report and opens it automatically when the run completes. The console output is useful. The HTML report is the client-ready deliverable.

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Environment	PowerShell 7.2 or later
Outputs	Colored console summary, CSV, and standalone HTML report

1. What the script does

This script answers a question that normally takes too much portal-hopping: what is consuming storage across a Microsoft 365 tenant? It discovers tenant information, measures Exchange Online primary and archive mailboxes, OneDrive, and SharePoint (including Teams-connected sites), and generates a console summary, CSV, and standalone HTML report. Backup cost estimates are optional.

- Exchange primary and shared mailbox storage from the Microsoft Graph mailbox usage report in Fast mode.
- Active Exchange Online archive mailbox storage, included by default.
- All OneDrive for Business personal-site storage.
- All SharePoint Online site storage, including Teams-connected sites.
- Teams files through their underlying SharePoint and OneDrive storage locations.

> **Scope note:** Teams chat and channel-message data is not separately calculated. Deleted or versioned content retained by Microsoft 365 Backup may increase actual billed usage.

2. First-time workstation setup

Use PowerShell 7. On a new workstation, install the required modules once. Future-you will appreciate not having to rediscover them mid-assessment:

```
Install-Module Microsoft.Graph.Authentication
Install-Module Microsoft.Graph.Sites
Install-Module Microsoft.Graph.Reports
Install-Module ExchangeOnlineManagement
Install-Module Microsoft.Online.SharePoint.PowerShell
```

Approve the trusted PSGallery and dependency prompts when appropriate for the administrative workstation. As always, read the prompt before enthusiastically pressing Y.

3. Before each run

1. Open a normal browser session, not InPrivate or Incognito.
2. Sign in to Microsoft 365 with the administrator account for the tenant being assessed and leave the browser open. Passkey authentication has been successfully used with the script.
3. Open PowerShell 7 and navigate to the folder containing the script.
4. Allow this unsigned script for the current PowerShell process only:

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

> Multiple sign-ins are expected: Microsoft Graph, Exchange Online, and SharePoint Online use separate authentication sessions. Complete each browser or device sign-in prompt and select the correct administrator account for the tenant. Several prompts do not necessarily mean a login failed.

4. Run the assessment

Default run: recommended

This is the normal run and the right starting point for most assessments. Fast Exchange mode is used by default, online archives are included unless you explicitly skip them, and the HTML report opens automatically when the work is finished.

```
.\M365-Tenant-Storage-Assessment-v2.4.ps1
```

Skip online archives

Use this when archive statistics are not required or when Exchange Online decides archive queries should become a character-building exercise. The HTML report clearly records that archives were skipped.

```
.\M365-Tenant-Storage-Assessment-v2.4.ps1 -SkipArchives
```

Do not automatically open the HTML report

```
.\M365-Tenant-Storage-Assessment-v2.4.ps1 -NoOpenReport
```

Exact Exchange mode

Exact mode queries mailbox statistics one mailbox at a time. It is slower—sometimes impressively so—but remains useful for comparison and troubleshooting.

```
.\M365-Tenant-Storage-Assessment-v2.4.ps1 -ExchangeMode Exact
```

Optional backup cost estimate

Backup pricing is hidden by default. To add estimated monthly and annual costs to the HTML report and console output, enable the optional switch. The rate is a planning assumption, not a provider quote.

.\M365-Tenant-Storage-Assessment-v2.4.ps1 -IncludeBackupEstimate -BackupPricePerGB 0.15

5. The HTML report: the primary deliverable

At the end of a successful run, the script writes a self-contained HTML report to the output folder and opens it in the default browser. You can review it locally, attach it to a ticket, archive it with client notes, or print it to PDF without hauling along a folder full of supporting files.

Treat the HTML report as the finished deliverable. The CSV is there for compact export and further analysis; the HTML report is what you use for technical review, client presentation, ticket documentation, archival, and PDF generation.

- Tenant identity, primary domain, SharePoint admin URL, and assessment timestamp.
- Total tenant data, with separate primary-mailbox and online-archive storage totals. Optional backup cost estimates are available when requested.
- Exchange Online, OneDrive, and SharePoint totals with each workload's percentage of the overall tenant storage estimate.
- A three-color stacked workload distribution bar: orange for Exchange, gold for OneDrive, and Microsoft blue for SharePoint.
- The Exchange measurement method and archive-collection outcome.
- A visible warning when one or more archive queries fail or time out.

> **Do not miss it:** The HTML file is the report intended for review and presentation. The CSV is the compact data export; the HTML is the polished assessment.

Figure 2. Illustrative storage dashboard with fictional tenant data. Backup pricing is shown only when -IncludeBackupEstimate is enabled.

● Exchange 81.8% ● OneDrive 11.9% ● SharePoint 6.3%

SERVICE	USED DATA	SHARE OF TOTAL
Exchange Online	3.14 TB	81.8%
OneDrive	467.21 GB	11.9%
SharePoint	248.52 GB	6.3%



TOTAL TENANT DATA	EXCHANGE ONLINE	ONEDRIVE + SHAREPOINT
3.84 TB	3.14 TB	717.73 GB

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Microsoft 365 Tenant Storage Assessment

Contoso Manufacturing | Generated October 7, 2026 5:37 PM

HTML assessment example

Figure 1. Illustrative console summary with fictional tenant data. The console now separates primary mailbox and online archive usage.

Console completion summary

The following examples use Contoso Manufacturing and synthetic identifiers. No client names, tenant IDs, domains, URLs, or report paths from production environments are shown.

Sample figures are illustrative; actual tenant totals will vary.

6. Exchange archive behavior

Online archives are included by default. After the Graph mailbox report finishes, the script queries each active archive mailbox individually. This is also the part where Exchange Online may choose to contemplate the meaning of time.

Real-world testing confirmed that Exchange Online archive-statistics requests may occasionally time out even when other archives complete quickly. When that occurs, the script continues processing and clearly identifies the archive collection as partial in both the console output and HTML report.

> **Microsoft black magic, professionally translated:** Some Exchange Online archive-statistics requests may take much longer than others or time out. A timeout does not stop the assessment. The script records the failure, continues processing, generates the reports, and warns that the Exchange total may be understated.

- Included: every discovered archive was measured successfully.
- Partial: one or more archives failed or timed out; the measured archive total is included, but the final Exchange total may be low.
- Skipped by administrator: the script was run with -SkipArchives.

If the same archive repeatedly times out, compare the Fast result with a prior Exact run or investigate that archive separately. Most importantly, do not let a partial result quietly put on a fake mustache and pass itself off as a complete live total.

7. Output location and files

By default, reports are written to:

```
$env:USERPROFILE\Documents\M365-Storage-Assessments
```

Each run creates:

- A timestamped CSV summary.

- A timestamped, standalone HTML assessment report.

8. Troubleshooting

Symptom	What to do
Script is not digitally signed	Run Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass in the current PowerShell 7 session.
SharePoint says no valid OAuth session exists	Confirm the script uses Connect-SPOService with -ModernAuth $true and -UseSystemBrowser $true, then complete the browser sign-in.
Several authentication prompts appear	Expected. Graph, Exchange, and SharePoint use separate sessions, especially when changing tenants.
An archive times out	The script continues and marks archive collection as partial. Use -SkipArchives when speed is more important than archive inclusion.
HTML does not open automatically	Open the displayed HTML path manually, or confirm the run was not started with -NoOpenReport.

9. Interpretation and limitations

Fast mode uses Microsoft Graph mailbox-usage report data for primary mailbox storage and live Exchange queries for archives. Graph usage data may lag live statistics. SharePoint and OneDrive figures depend on reported site storage. The assessment is a point-in-time estimate, not a complete inventory of every Microsoft 365 data type or a guaranteed backup invoice.

- Backup pricing is optional: use -IncludeBackupEstimate to display it and -BackupPricePerGB to change the planning rate.
- Deleted content, version history, and backup-retained recovery data can increase billed usage.
- A partial or skipped archive result understates Exchange storage by the omitted archive amount.
- Validate unusual results before using them in a proposal or client commitment.

10. Ownership and sharing

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> **Field note:** This tool exists because opening several admin portals, exporting separate reports, and manually stitching the totals together is a terrible use of an administrator's afternoon. The finished HTML report turns that work into a clear, client-ready assessment.